9. A piece of brass undergoes three different processes involving change of energy.

P: It is lifted vertically 2 m

Q: it is heated from 15°C to 20°C

R: It is accelerated from rest to 10 m s^{-1}

The specific heat capacity of brass is 380 $J^{-1}K^{-1}kg^{-1}$.

These processes, arranged in order of increasing energy change are

A *P Q R* **B** *Q P R* **C** *Q R P* **D** *P R Q*

Ans : D

$\Delta E_p = m\ g\ h = m\ (9.81)(2)$

$\Delta E_Q = m\ c\ \Delta\theta = m(380)(5)$

$\Delta E_R = \frac{1}{2} m\ v^2 - 0 = \frac{1}{2} m\ (100)$